

Lecture in Labour Economics

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Learning Objectives

1. Using the labour supply model, analyse the work incentive effects of the following policies:
 - a Unemployment insurance benefits
 - b Child care benefit
2. Explain how imperfect information can lead to a labour market equilibrium in which unemployed workers and unfilled job vacancies co-exist

Unemployment Insurance Benefits

- ▶ Canada's unemployment insurance program, known as Employment Insurance EI since 1996, is the largest income security program for non-elderly individuals in the country
- ▶ The amount of income replacement rate is 55% of lost earning, subject to a maximum
- ▶ The duration of benefit ranges from 14 to 45 weeks, depending on the regional rate of unemployment
- ▶ In order to qualify for the benefits, individuals must have worked between 420 and 700 hours, the equivalent of 12-20 weeks at 35 hours per week, again depending on the unemployment rate of the region.

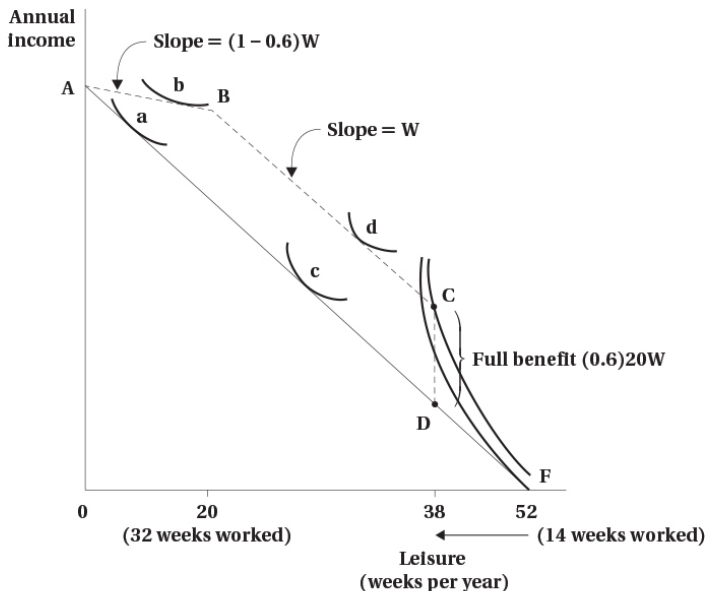
Unemployment Insurance Benefits

Example

Analysing the work incentive effects of unemployment insurance using the income-leisure model of the labour supply

- ▶ Recipients receive 60 percent of their weekly pay in the case of unemployment
- ▶ It requires a minimum of 14 weeks of employment in order to qualify for benefits.
- ▶ The maximum period of receiving benefits is 20 weeks
- ▶ For simplicity, we use one-year time horizon

Income Constraint for a Simplified Unemployment Insurance Scheme



Income Constraint for a Simplified Unemployment Insurance Scheme

- ▶ AF (solid line)- potential income constraint in the absence of unemployment insurance (UI) program
- ▶ ABCDF- constraint with UI
- ▶ If the individual works less than 14 weeks during the year, they do not qualify for UI benefits - their income constraint is segment DF
- ▶ If the individual works 14 weeks, they are eligible for benefits for an additional 20 weeks
- ▶ Their income at 14 weeks of work equals their labour market earning for the 14 weeks plus 20 times 60 percent of their normal weekly wage earnings

Income Constraint for a Simplified Unemployment Insurance Scheme

- ▶ Their annual income would be $Y = 14W + (0.60)20W$
 - ▶ W is weekly wage earnings
 - ▶ $(0.60)20W$ - full benefit (vertical distance between C and D)

The work incentive effects of UI depend on where the individual would be located in the absence of UI system - three different possibilities

1. Individuals working more than 32 weeks (e.g. point a on AF)
 - ▶ The income and substitution effects work in the same direction to potentially decrease weeks worked (moving to point b on AB)
2. Individuals such as those originally at point c on AF
 - ▶ the UI system constitutes pure income effect - potentially decreasing work incentives (moving to point d on AB)

Income Constraint for a Simplified Unemployment Insurance Scheme

3. Individuals who would be outside the labour force in the absence of UI program
 - ▶ originally located at point F - reflecting the high value of their time devoted to non-market activities
 - ▶ many of these individuals neither work nor collect UI
 - ▶ others would enter the labour force and work a sufficient number of weeks to qualify for UI benefits (point C)
 - ▶ UI program has increased work incentives by making it attractive for these individuals to devote less time to non-market activities and to enter the labour force for relatively brief periods.

Child Care Benefit

- ▶ Analysis of the child care benefit first requires an investigation of how child care costs might affect a parent's labour supply decision
- ▶ Child care costs can be modelled as a special case of fixed costs associated with working
- ▶ Assumptions:
 - ▶ child care cost is incurred only if the person works in paid labour market
 - ▶ child care cost is independent of hours worked

The effect of Child Care Costs on Labour Supply

Income

A

B

Y

$Y - D$

0

(a) The impact on
budget constraint

Cost of daycare

E

M

T

Leisure

The effect of Child Care Costs on Labour Supply

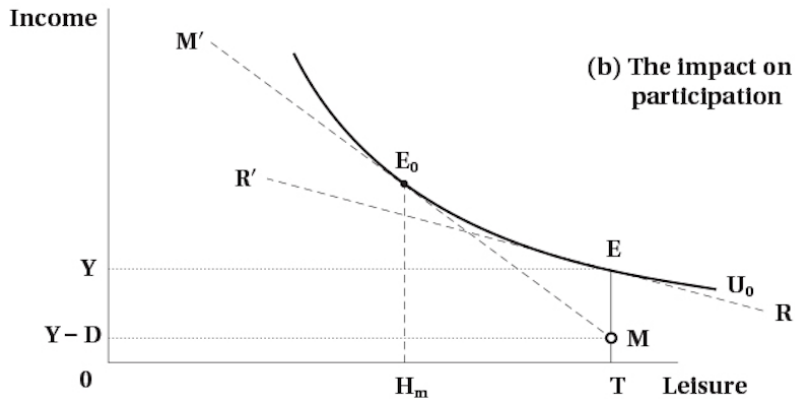
The impact on budget constraint:

- ▶ the fixed cost of child care is analogous to a vertical drop in the individual's budget constraint immediately upon entering the labour market
- ▶ the individual incurs the fixed costs $D=EM$ → analogous to a drop in income of EM
- ▶ point M is slightly to the left of the vertical line ET - indicating that costs are incurred only if one engages in labour market activity
- ▶ fixed costs are avoided if one does not work in the labour market but remains at E

Summary:

- ▶ Income constraint AE - in the absence of child care costs
- ▶ Income constraint BME - with fixed child care costs

The effect of Child Care Costs on Labour Supply



The effect of Child Care Costs on Labour Supply

The impact on participation:

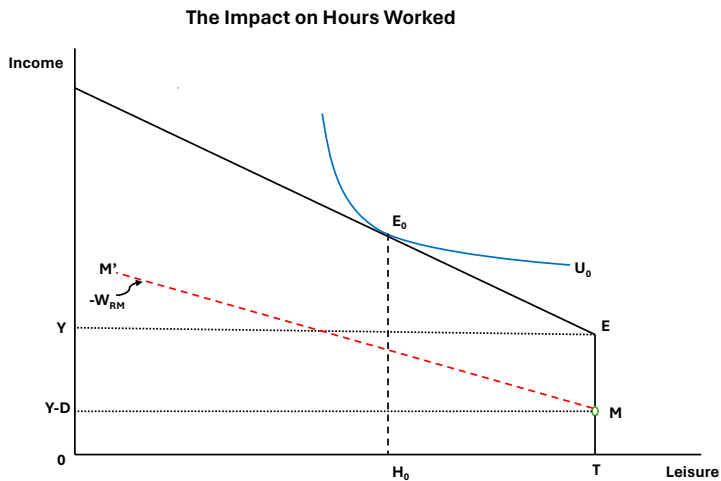
- ▶ In the absence of child care fixed costs, the individual's reservation wage is depicted by the slope RR'
- ▶ In the presence of child care fixed costs, the individual's reservation wage is depicted by the slope MM'
 - ▶ the minimum wage above which one would enter the labour market
 - ▶ at this wage one would be indifferent between engaging in labour market activities at E_0 and remaining at home with children at E
- ▶ Market wage greater than the reservation wage would induce labour force participation and the amount of work would be determined by the tangency of new, higher indifference curve and a higher budget constraint

The effect of Child Care Costs on Labour Supply

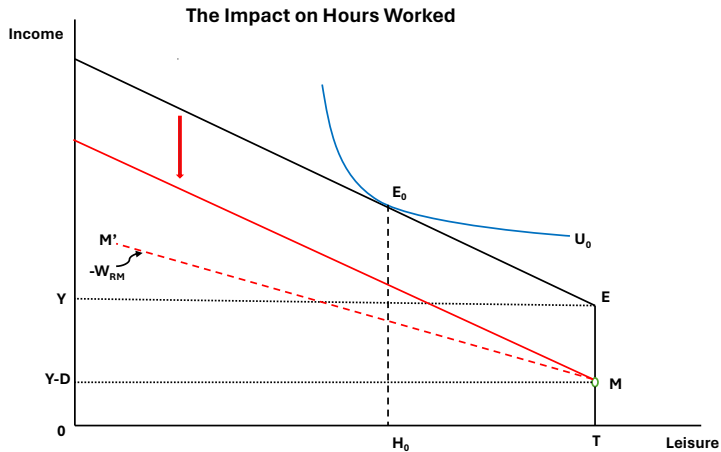
The impact on participation:

- ▶ the line MM' is tangent to U_0 at the point E_0 that is to the left of E of the same indifference curve
- ▶ the indifference curve become steeper as one moves to the left of E because of the hypothesis of diminishing marginal rate of substitution
- ▶ child care costs associated with entering the labour market increase the reservation wage by making labour market work less attractive than caring for the children at home

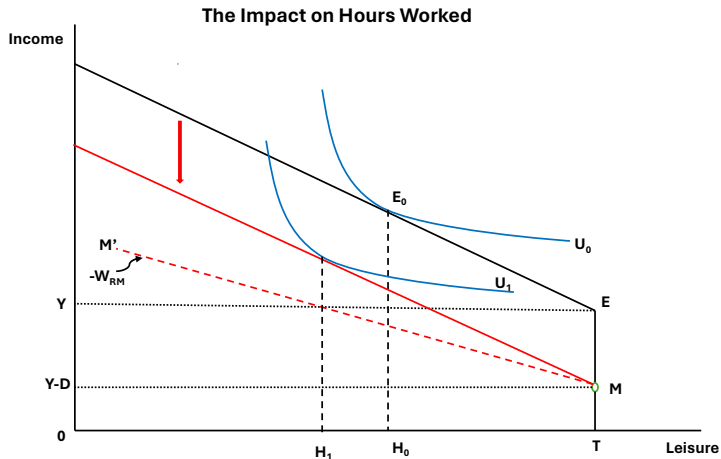
The effect of Child Care Costs on Labour Supply



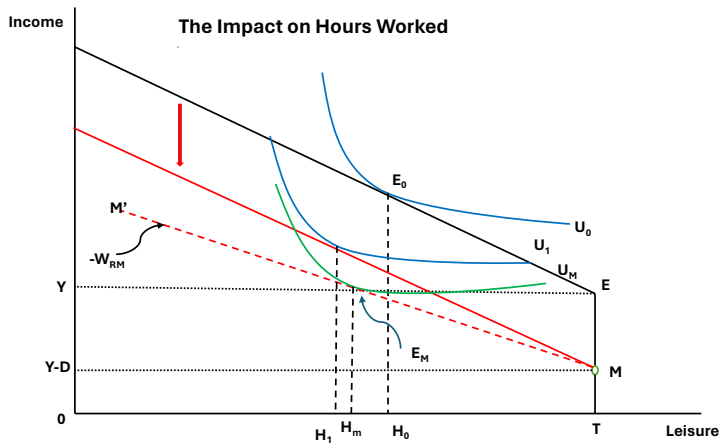
The effect of Child Care Costs on Labour Supply



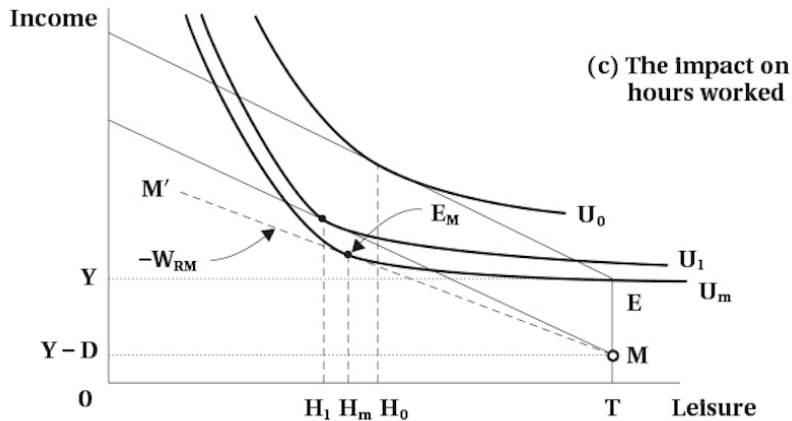
The effect of Child Care Costs on Labour Supply



The effect of Child Care Costs on Labour Supply



The effect of Child Care Costs on Labour Supply



The effect of Child Care Costs on Labour Supply

The impact on hours worked:

- ▶ initial equilibrium: at E_0 with hours of work H_0 and no child care costs
- ▶ the existence of fixed child care costs shifts the budget constraint down in a parallel fashion because market wages have not changed
- ▶ if leisure is a normal good - this pure income effect from the reduced income increases hours of work from H_0 to H_1
- ▶ when there are fixed costs, one has to work more hours to make up for the income loss associated with the costs of child care.

The effect of Child Care Costs on Labour Supply

Fixed child care costs also create a discontinuity or gap in the implied labour supply schedule

- ▶ the distance TH_m indicates the number of hours below which it would not be worthwhile to enter the labour market - given the reservation wage W_{RM} created by the child care costs EM
- ▶ it is not possible to amortize the fixed costs of entering the labour force over so few hours $\rightarrow$ there is a discrete jump in the labour supply from zero to TH_m
- ▶ geometrically, it is not possible for there to be a tangency between the budget constraint MM' and the indifference curve U_M in the region E_ME
 - ▶ the slope of MM' is always greater than the slope of the indifference curve to the right E_M
 - ▶ in this way, the child care costs discourage part-time work

Child Care Benefit

- ▶ Effects of child care costs
 - ▶ Fixed child care costs increase reservation wage - decrease labour force participation
 - ▶ Fixed child care costs also affect hours-of-work
 - ▶ Increased hours for those who participate
 - ▶ Create a discontinuity in labour supply - making part-time work less likely

Therefore, child care benefit:

- ▶ encourages labour force participation and part-time work
- ▶ reduces the hours of work for those already participating

Imperfect Information in the Labour Market

- ▶ Imperfect information in the labour market can result in a situation where both job vacancies and unemployed workers exist simultaneously
- ▶ frictional unemployment - unemployment associated with normal turnover in the labour market
- ▶ structural unemployment - mismatch between labour demand and labour supply, either on occupational, skills, or regional dimensions

Imperfect Information in the Labour Market

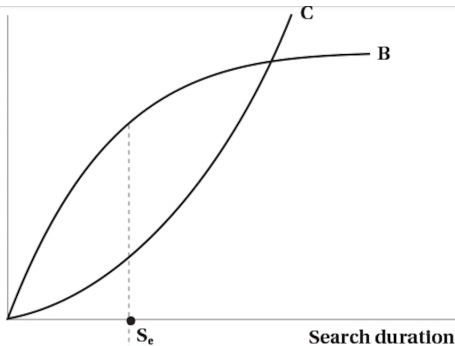
- ▶ the job search and matching process is associated with imperfect information on both sides of the labour market
- ▶ unemployed workers are not aware of
 - ▶ all available jobs
 - ▶ their rates of pay
 - ▶ location
 - ▶ working condition
- ▶ employers with job vacancies are not aware of all individual workers and their characteristics
- ▶ if both sides were fully informed, the process of matching workers and jobs could take place in a few days, if not hours.
- ▶ since the acquisition of information about job opportunities and job applicants takes time, unemployment and vacancies co-exist.

Imperfect Information in the Labour Market

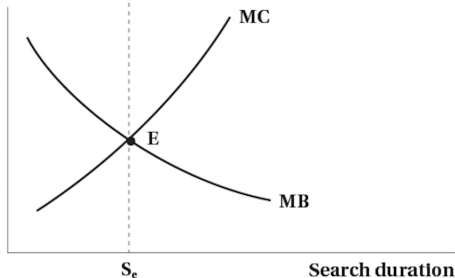
- ▶ Job search is an economic decision in that it involves both costs and benefits
- ▶ two main aspects to the decision:
 - ▶ determining whether it is worthwhile initiating the job search process
 - ▶ determining when to stop the process
- ▶ Expected benefits have to be weighted against the expected costs
- ▶ Employees continue to search until expected marginal benefit equal expected marginal cost
- ▶ Benefits and costs of search are likely to be related to the amount of time devoted to a job search
- ▶ Marginal cost is rising with the amount of job search undertaken

Optimal Job Search

Present value
of total costs
(C) and
expected total
benefits (B)



Present value
of marginal
costs (MC)
and marginal
expected
benefits (MB)



Imperfect Information in the Labour Market

- ▶ Optimal amount of a search will maximize the difference between B and C
- ▶ The difference is maximized when expected marginal benefit equals expected marginal cost at S_e

Imperfect Information in the Labour Market

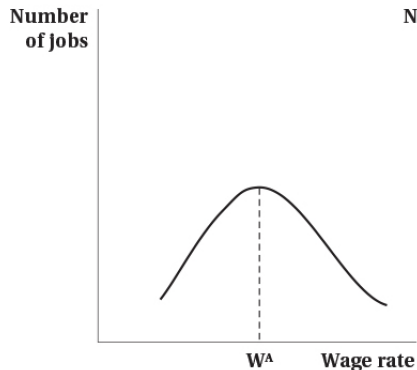
Implications of imperfect information

Generally workers and firms will stop their search activities prior to being fully informed due to diminishing returns to information acquisition and also increasing costs of information acquisition. As a result:

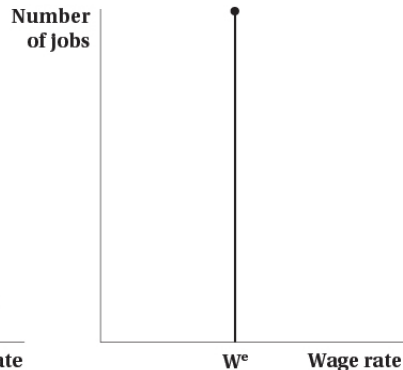
- ▶ A labour market characterized by imperfect information will not clear instantaneously
- ▶ Dispersion of wage rates will occur even in a labour market with homogeneous workers and jobs

Wage Distribution Under Imperfect and Perfect Information

(a) Imperfect Information



(b) Full Information



Imperfect Information in the Labour Market

Example:

Job Search and the Reservation Wage

Julia just graduated from college and is looking for a first job.
There are three types of jobs:

- ▶ The best jobs pay \$2400 a month
- ▶ “Not as good” jobs pay \$2200 a month
- ▶ The worst type of job pays \$2000 a month
- ▶ The probability of each type of job offered is one-third

Imperfect Information in the Labour Market

Julia must now decide what to do once she gets an offer

- ▶ The expected value of a future offer is \$2200 a month (average of \$2000, \$2200 , and \$2400 a month)
- ▶ She won't accept a job offer unless it pays at least \$2200 a month
- ▶ $\frac{2}{3}$ is the hazard rate

Imperfect Information in the Labour Market

Factors Determining Optimal Search

- ▶ Expected benefit of job search
- ▶ Expected duration of a job
- ▶ Age: younger workers' unemployment duration is lower
- ▶ Internet accessibility
- ▶ Number of employers with available job vacancies
- ▶ Rate at which employers offer jobs
- ▶ Value of leisure
- ▶ Social and labour market policies
- ▶ Employment insurance
- ▶ Pensions

Reference

Labour Market Economics, 9th edition, by Dwayne Benjamin, Morley Gunderson, Thomas Lemieux, Craig Riddell, Tammy Schirle, 2021, McGraw-Hill Ryerson.

Appendix: Basic Income-Leisure Model

The choice of hours worked given opportunities and value of non-market time

- ▶ Preferences and Constraints
- ▶ Individuals choose the feasible outcomes which yield the highest level of satisfaction

Preferences

Two “goods”

- ▶ Consumption
- ▶ Leisure

Represented by indifference curves

(A person is indifferent between various combinations of consumption and leisure on an indifference curve)

Appendix: Basic Income-Leisure Model

Preferences

- ▶ We assume that consumers have well-defined preferences over all the conceivable combinations of consumption and leisure
- ▶ This implies that all combinations lie on some indifference curve
- ▶ Consumer preferences can then be represented by an indifference curve
- ▶ Indifference curves plot combinations of consumption and leisure that yield the consumer equal levels of utility
- ▶ The absolute value of the slope of an indifference curve gives the marginal rate of substitution (MRS); that is, the amount of leisure the consumer is willing to accept in exchange for giving up some consumption

Appendix: Basic Income-Leisure Model

Constraint

- ▶ Constraints are determined by the economic properties of the market, which, in turn, transform consumption-leisure to income-leisure by setting the price of consumption
- ▶ The income and leisure combinations from which the consumer can choose with a potential income constraint
- ▶ This allows a transformation of our model from representing consumption-leisure tradeoffs to income-leisure tradeoffs.
- ▶ Alternative phrases used for the potential income constraint include budget constraint, income constraint, and full-income constraint

Appendix: Basic Income-Leisure Model

The Consumer's Optimum

- ▶ Optimal amount of income and leisure

Utility-maximizing equilibrium

- ▶ Highest indifference curve given the income constraint

Compare MRS with the Market Wage Rate

- ▶ MRS: measures the willingness to exchange leisure for consumption (or income)
- ▶ Market Wage Rate: measures the ability to exchange leisure for income

Appendix: Basic Income-Leisure Model

Change in Wage Rate on Hours of Work

Two effects:

1. **Income effect** The worker has more income to buy more goods including leisure (reduces work hours)
2. **Substitution effect** Individual may work more because the returns are greater substituting away from leisure